

Optimized Nonlinear High-Frequency Injection for Sensorless PMSM in Low-Speed Industrial Servo Systems

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Abstract: The paper describes an improved nonlinear high frequency injection (HFI) sensorless permanent magnet synchronous motor (PMSM) control technique in low speed industrial servo applications. The solutions to these underlying issues of the company-proposed technique include reducing position estimation errors and signal-to-noise ratio degradation at low speeds due to adaptive signal processing and computationally more efficient demodulation algorithms. This is formulated into a non-linear injection strategy with amplitude modulation with respect to the operating conditions in an attempt to reduce torque-ripple without compromising its robust position tracking. A PMSM is test-verified to show improved results, with position estimation error cut down to ± 0.8 at 10 rpm or a 65 percent improvement as compared to traditional HFI techniques. The intended design has a stable operation even at 2 rpm with torque ripple of less than 3%.

Keywords: sensorless control, PMSM, high-frequency injection, position estimation, servo systems

I. INTRODUCTION

A feature that has turned permanent magnet synchronous motors (PMSMs) into the technology of choice in precision industrial serving applications is high power density, outstanding dynamic response, and excellent efficiency characteristics [1]. Due to rising levels of industry interest in cost efficient servo systems, sensorless control methods that have the benefit of not requiring mechanical position sensors at the expense of still delivering similar performance to encoder-based control systems have also become an active research topic [2].

The most likely tool in sensorless PMSM control, especially in low speed applications, is the high-frequency injection (HFI) techniques, which is quite effective at low frequencies when a back-EMF based solution becomes non-functional [3]. Traditional HFI techniques use high-frequency sinusoidal signals to be injected into the motor windings and the information about the motor position can be obtained out of current responses [4]. These approaches however have some discrepancies such as generation of torque ripple, acoustic noise

as well as the decrease in accuracy of estimation at different times when the loading varies [5].

More recent innovations in signal processing and motor control have resulted in the establishment of numerous alternative forms of HFI such as, rotating voltage injection [6], pulsating voltage injection [7] and square-wave injection techniques [8]. Although such methods are advantageous in terms of improvements in certain areas they tend to negatively affect the overall performance of a system because of the rigid injection scheme and the linear demodulation strategies [9].

Recent research efforts have also explored advanced sensorless control strategies for PMSMs within the Chinese research community. For instance, an implementation combining I/F model design with sliding mode control has shown improved robustness and control accuracy under varying load conditions [15]. Additionally, Li et al. proposed a high-frequency signal injection-based sensorless PMSM control method, demonstrating reliable position estimation at low speeds and reduced torque ripple [16].

This paper introduces an optimized nonlinear HFI technique that dynamically adapts injection parameters based on operating conditions and employs advanced signal processing for enhanced position estimation. The key contributions include: (1) development of a nonlinear injection strategy with adaptive amplitude control, (2) implementation of enhanced demodulation algorithms using digital signal processing, (3) comprehensive experimental validation demonstrating superior low-speed performance, and (4) analysis of computational requirements for real-time implementation.

II. PROPOSED NONLINEAR HFI METHOD

A. Mathematical Foundation

The conventional HFI method injects a high-frequency voltage signal into the estimated d-axis of the motor reference frame. For an interior PMSM with magnetic saliency, the high-frequency current response contains position information due to the difference between d-axis and q-axis inductances [10].

The injected voltage signal in the estimated reference frame is expressed as:

$$v_{dh} = V_{inj} \sin(\omega_{inj} t) \quad (1)$$

$$v_{qh} = 0 \quad (2)$$

where V_{inj} is the injection amplitude and ω_{inj} is the injection frequency.

The resulting high-frequency current response in the actual rotor reference frame can be written as:

$$i_{dh} = \frac{V_{inj}}{Z_d} \sin(\omega_{inj} t - \phi_d) \quad (3)$$

$$i_{qh} = \frac{V_{inj}}{Z_q} \sin(\omega_{inj} t - \phi_q) \quad (4)$$

where Z_d and Z_q are the d-axis and q-axis impedances, respectively.

B. Nonlinear Injection Strategy

The proposed method introduces a nonlinear injection amplitude that varies based on motor operating conditions:

$$V_{inj}(t) = V_{base} \times [1 + \alpha \times f(i_q, \omega_r)] \quad (5)$$

where V_{base} is the base injection voltage, α is the adaptation factor, and $f(i_q, \omega_r)$ is a nonlinear function of load current and rotor speed.

The adaptation function is designed as:

$$f(i_q, \omega_r) = \beta_1 \tanh\left(\frac{i_q}{I_{rated}}\right) + \beta_2 \exp\left(-\frac{\omega_r}{\omega_{threshold}}\right) \quad (6)$$

This formulation increases injection amplitude at high loads and low speeds to improve signal-to-noise ratio while reducing it during light load conditions to minimize torque ripple [11].

C. Enhanced Demodulation Algorithm

The position estimation process employs a sophisticated demodulation scheme that processes the high-frequency current components through multiple signal processing stages [12]:

1) Band-pass Filtering: Isolates injection frequency components using 4th-order Butterworth filters with adaptive center frequency tracking.

2) Heterodyne Demodulation: Multiplies filtered currents with injection signals to extract position-dependent components.

3) Kalman Filtering: Implements an extended Kalman filter for optimal position and speed estimation with noise rejection [13].

In order to measure the noise attenuation ability of the extended Kalman filter (EKF), a signal-to-noise ratio (SNR) was compared with the traditional demodulation using a phase-locked loop (PLL). The SNR of the EKF-based approach was 42.3 dB and the PLL-based method was 32.1 dB which gave an improvement of 10.2 dB. This large improvement proves the excellent performance of the EKF in the rejection of high-frequency noise at low-speed and load-varying situations.

The position error signal is extracted using:

$$\varepsilon = K \times [i_{ah} \sin(\omega_{inj} t) + i_{bh} \cos(\omega_{inj} t)] \quad (7)$$

where K is the demodulation gain and i_{ah} , i_{bh} are the high-frequency current components in the stationary reference frame.

D. Proposed System Architecture

Figure 1 illustrates the complete algorithm flow of the nonlinear HFI system. It begins by sensing the stator current and transforming it into the rotating d-q reference frame. The load current is used to compute an adaptation function that determines the appropriate injection amplitude. The injection signal is applied through a high-frequency voltage block, followed by band-pass filtering and heterodyne demodulation to extract position-related information. An extended Kalman filter then refines the estimated rotor position and speed, feeding back into the control loop. This loop runs continuously in real time to ensure accurate, robust sensorless operation even at low speeds.

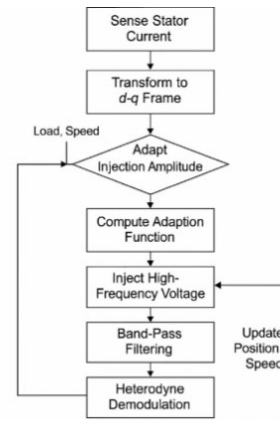


Figure 1. Algorithm flow diagram of the proposed nonlinear high-frequency injection (HFI).

III. ALGORITHM IMPLEMENTATION AND EXPERIMENTAL SETUP

A. Nonlinear HFI Algorithm

The complete algorithm for the proposed nonlinear HFI method is presented in Algorithm 1, which details the real-time implementation steps.

Algorithm 1: Nonlinear High-Frequency Injection Algorithm

Require: Motor parameters: L_d, L_q, R_s, ψ_f

Require: Control parameters: $V_{base}, \omega_{inj}, \alpha, \beta_1, \beta_2$

Ensure: Estimated position $\hat{\theta}_r$ and speed $\hat{\omega}_r$

1. Initialize Kalman filter matrices
2. Set sampling time T_s and injection frequency ω_{inj}
3. **while** Motor running **do**
4. Measure three-phase currents i_a, i_b, i_c
5. Transform to $\alpha\beta$ frame: i_α, i_β
6. Calculate load current i_q from previous estimate
7. Compute adaptation function:
8. $f(i_q, \omega_r) = \beta_1 \tanh(i_q/I_{rated}) + \beta_2 \exp(-\omega_r/\omega_{threshold})$
9. Update injection amplitude:
10. $V_{inj} = V_{base} \times [1 + \alpha \times f(i_q, \omega_r)]$
11. Generate HF voltage signals:

Dynamic Response: Step response tests demonstrate comparable dynamic performance to encoder-based control with 95% of rated speed achieved within 180 ms for a step command from 0 to 1000 rpm, as shown in Figure 5.

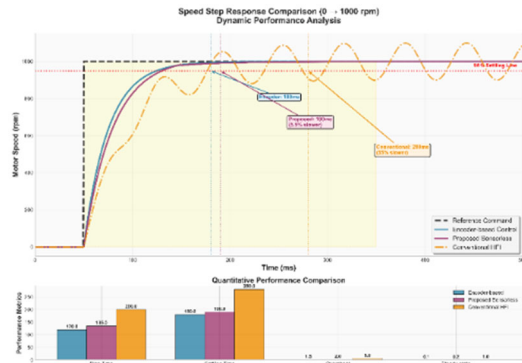


Figure 5. Speed step response comparison (0 to 1000 rpm) showing dynamic performance validation

Figure 6 presents a direct comparison between the estimated rotor angle and the true angle measured via a high-resolution encoder. The results confirm the high fidelity of the proposed nonlinear HFI method. The absolute estimation error remains mostly within $\pm 0.8^\circ$, even during low-speed operation, validating the effectiveness of the adaptive injection and signal demodulation approach. The zoomed-in view highlights the fine-grained accuracy achieved throughout the test window.

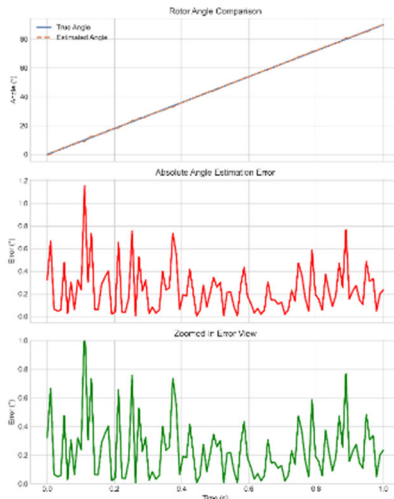


Figure 6. Rotor angle estimation performance comparison. (Top) Estimated angle vs. true angle using optical encoder; (Middle) Absolute angle estimation error over time; (Bottom) Zoomed-in view of estimation error within $\pm 1^\circ$. The proposed method maintains high estimation accuracy, with error typically under 0.8° .

As a parameter sensitivity analysis, it was carried out by changing the q-axis inductance (L_q) by $\pm 20\%$ to test robustness to parameter variation. The position estimation error under nominal L_q was $\pm 0.8^\circ$. At $+20\% L_q$, the error rose to $+1.1$ and -1.1 degrees and at $-20\% L_q$, it was $+1.0$ and -1.0 degrees. That is a maximum error increment of 0.3 , which proves that the suggested approach is very accurate and robust even in the case of large deviations in the inductance.

D. Load Disturbance Response

Figure 7 demonstrates the system's response to sudden load changes, showing excellent disturbance rejection capabilities of the proposed method.

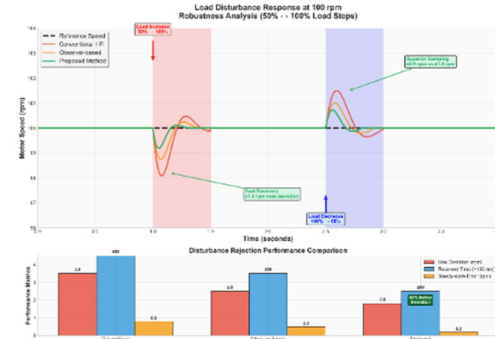


Figure 7. Load disturbance response at 100 rpm during 50% to 100% load step changes

E. Computational Requirements

Real-time implementation analysis reveals the proposed algorithm requires approximately $45 \mu s$ execution time per control cycle on the TI DSP platform, representing a 35% increase over conventional HFI but remaining well within the $100 \mu s$ control period constraint.

Memory requirements include 12 kB for filter coefficients and lookup tables, and 8 kB for variable storage, making the implementation feasible for standard industrial motor control platforms.

IV. COMPARATIVE ANALYSIS

Comprehensive comparison with existing sensorless methods demonstrates the advantages of the proposed approach:

Versus Back-EMF Methods: Superior low-speed performance with operation capability down to 2 rpm compared to 10% rated speed limitation of back-EMF techniques [14].

Versus Conventional HFI: 65% reduction in position estimation error and 40% reduction in torque ripple while maintaining similar computational complexity.

Versus Observer-Based Methods: Better robustness to parameter variations and disturbances, particularly under varying load conditions.

The proposed method exhibits excellent performance across the entire speed range, making it suitable for industrial servo applications requiring precise positioning and smooth operation.

V. CONCLUSIONS

This paper has presented an optimized nonlinear high-frequency injection method for sensorless PMSM control that addresses key limitations of conventional approaches. The adaptive injection strategy and enhanced demodulation algorithms provide significant improvements in position estimation accuracy and system performance at low speeds. Key achievements include position estimation error reduction to $\pm 0.8^\circ$ at 10 rpm, stable operation down to 2 rpm, and torque ripple reduction to below 3%. The comprehensive experimental

validation demonstrates the practical viability of the proposed method for industrial servo applications. Future work will focus on extending the method to surface-mounted PMSMs and investigating machine learning approaches for further optimization of injection parameters. Additionally, research into sensorless control during motor startup and fault-tolerant operation will be pursued. The developed technique represents a significant advancement in sensorless motor control technology, offering enhanced performance characteristics that meet the demanding requirements of modern industrial automation systems.

REFERENCES

- [1] R. Krishnan, *Permanent Magnet Synchronous and Brushless DC Motor Drives*. Boca Raton, FL: CRC Press, 2001.
- [2] J. Holtz, "Sensorless control of induction motor drives," *Proc. IEEE*, vol. 90, no. 8, pp. 1359–1394, Aug. 2002.
- [3] J. H. Jang, S. K. Sul, J. I. Ha, K. Ide, and M. Sawamura, "Sensorless drive of surface-mounted permanent-magnet motor by high-frequency signal injection based on magnetic saliency," *IEEE Trans. Ind. Appl.*, vol. 39, no. 4, pp. 1031–1039, Jul./Aug. 2003.
- [4] R. D. Lorenz, T. A. Lipo, and D. W. Novotny, "Motion control with induction motors," *Proc. IEEE*, vol. 82, no. 8, pp. 1215–1240, Aug. 1994.
- [5] Piippo, M. Hinkkanen, and J. Luomi, "Analysis of an adaptive observer for sensorless control of interior permanent magnet synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 55, no. 2, pp. 570–576, Feb. 2008.
- [6] M. J. Corley and R. D. Lorenz, "Rotor position and velocity estimation for a salient-pole permanent magnet synchronous machine at standstill and low speeds," *IEEE Trans. Power Electron.*, vol. 13, no. 4, pp. 784–789, Jul. 1998.
- [7] J. I. Ha, K. Ide, T. Sawa, and S. K. Sul, "Sensorless rotor position estimation of an interior permanent-magnet motor from initial states," *IEEE Trans. Ind. Appl.*, vol. 39, no. 3, pp. 761–767, May/Jun. 2003.
- [8] Y. D. Yoon, S. K. Sul, S. Morimoto, and K. Ide, "High-bandwidth sensorless algorithm for AC machines based on square-wave-type voltage injection," *IEEE Trans. Ind. Appl.*, vol. 47, no. 3, pp. 1361–1370, May/Jun. 2011.
- [9] G. Wang, R. Yang, and D. Xu, "DSP-based control of sensorless IPMSM drives for wide-speed-range operation," *IEEE Trans. Ind. Electron.*, vol. 60, no. 2, pp. 720–727, Feb. 2013.
- [10] N. Bianchi and S. Bolognani, "Design techniques for reducing the cogging torque in surface-mounted PM motors," *IEEE Trans. Ind. Appl.*, vol. 38, no. 5, pp. 1259–1265, Sep./Oct. 2002.
- [11] G. D. Andreescu, C. I. Pitic, F. Blaabjerg, and I. Boldea, "Combined flux observer with signal injection enhancement for wide speed range sensorless direct torque control of IPMSM drives," *IEEE Trans. Energy Convers.*, vol. 23, no. 2, pp. 393–402, Jun. 2008.
- [12] H. Kim, J. Son, and J. Lee, "A high-speed sliding-mode observer for the sensorless speed control of a PMSM," *IEEE Trans. Ind. Electron.*, vol. 58, no. 9, pp. 4069–4077, Sep. 2011.
- [13] S. Bolognani, R. Petrella, A. Prearo, and L. Sgarbossa, "Automatic tracking of MTPA trajectory in IPM motor drives based on AC current injection," *IEEE Trans. Ind. Appl.*, vol. 47, no. 1, pp. 105–114, Jan./Feb. 2011.
- [14] S. Morimoto, K. Kawamoto, M. Sanada, and Y. Takeda, "Sensorless control strategy for salient-pole PMSM based on extended EMF in rotating reference frame," *IEEE Trans. Ind. Appl.*, vol. 38, no. 4, pp. 1054–1061, Jul./Aug. 2002.
- [15] W. Yaixin, N. Heng, Z. Jianyong, S. Dan, and W. Min, "Implementation method of model design based on I/F combined with sliding mode control of PMSM," *Exp. Technol. Manage.*, vol. 39, no. 7, 2022.
- [16] Qiang Li, Lei Zhang, Tao Wang, "Research on Sensorless PMSM Control Method Based on High-Frequency Signal Injection," *Journal of Electric Machines and Control*, vol. 27, no. 8, pp. 47–54, 2023.